

Implementation Guide – PUBLIC
Version 2.3 (2026-05-16)

How to Integrate with SAP Business Network Freight Collaboration (ANSI X12)

Implementation guide for B2B EDI ANSI X12 integration

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INTRODUCTION

Note: As of today, SAP Business Network Freight Collaboration supports the following B2B standards and corresponding messages:

EDI/B2B Message Standard	Message Type	Version	Description
EDI	ANSI 204	V4010	Load tender, submitting shipment to carriers
EDI	ANSI 214	V4010	Shipment status
EDI	ANSI 300	V4010	Freight ocean booking request
EDI	ANSI 301	V4010	Freight booking confirmation
EDI	ANSI 303	V4010	Booking cancellation (ocean)
EDI	ANSI 310	V4010	Freight receipt and invoice (ocean)
EDI	ANSI 315	V4010	Status details (ocean)
EDI	ANSI 840	V4010	Request for Quotation
EDI	ANSI 843	V4010	Response to Request for Quotation
EDI	ANSI 864	V4010	Text Message (Quotation Notification)
EDI	ANSI 990	V4010	Response to “Load tender”
EDI	ANSI 210	V4010	Motor carrier freight details and invoice
EDI	ANSI 997	V4010	Functional acknowledgment

To set up ANSI X12 integration, an SFTP connection needs to be configured in the *System Connections* app. This implementation guide describes in detail what you need to do.

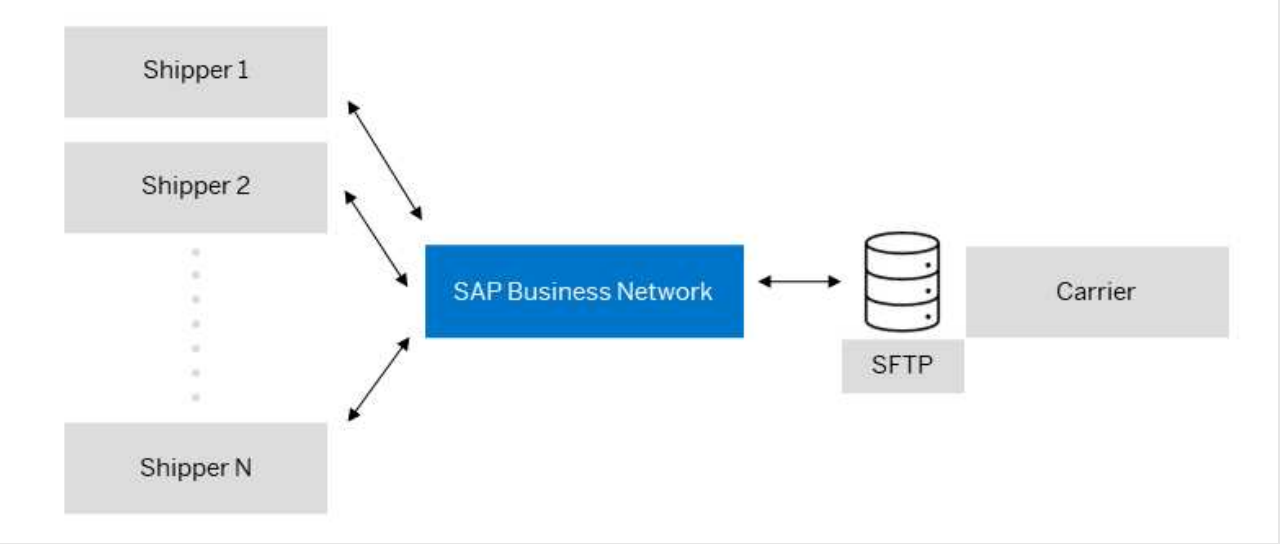
Note: All activities mentioned in this guide have to be done in both the test and productive environments.

PREREQUISITES

To set up EDI integration with SAP Business Network for Logistics, you must meet the following requirements:

- Carrier tenant is available in the test and productive environments. For more information about the process of registering SAP Business Network for Logistics, activating your account, logging on, and adding users, see the Activate Your Account guide.
- In case your SFTP server uses IP filtering, you need to add specific IP ranges to the allow list based on the data center you're onboarded to. Otherwise, the Cloud Integration tenants cannot connect to your SFTP server. For more information, see Regions and Hosts Available for the Connectivity Platform | SAP Help Portal.
- It is expected that a carrier sets up an SFTP server on their side for information exchange. Carriers have to set up the SFTP connection once and shippers can use the same SFTP connection for communication. Carriers must enter a base64-encoded string of their SFTP server in the *Known Host Key* field of the *System Connections* app. See also Specifying the Authentication Details and Base URLs for Your System's Endpoints.

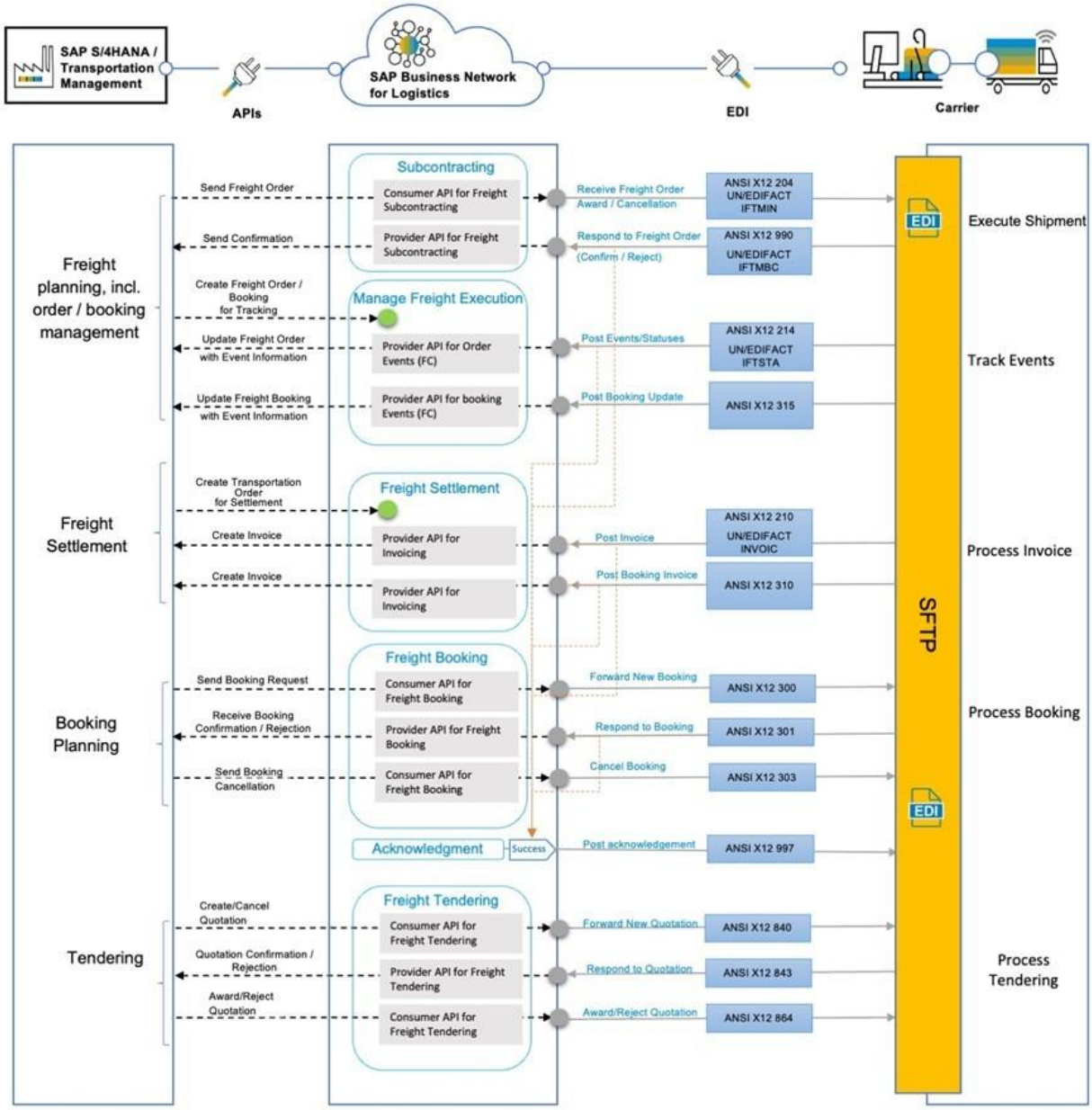
  The following diagram outlines the SFTP communication between the involved parties:



- Ensure that you use different folders for *Test* and *Prod* on your SFTP server.

PROCESS DESCRIPTION

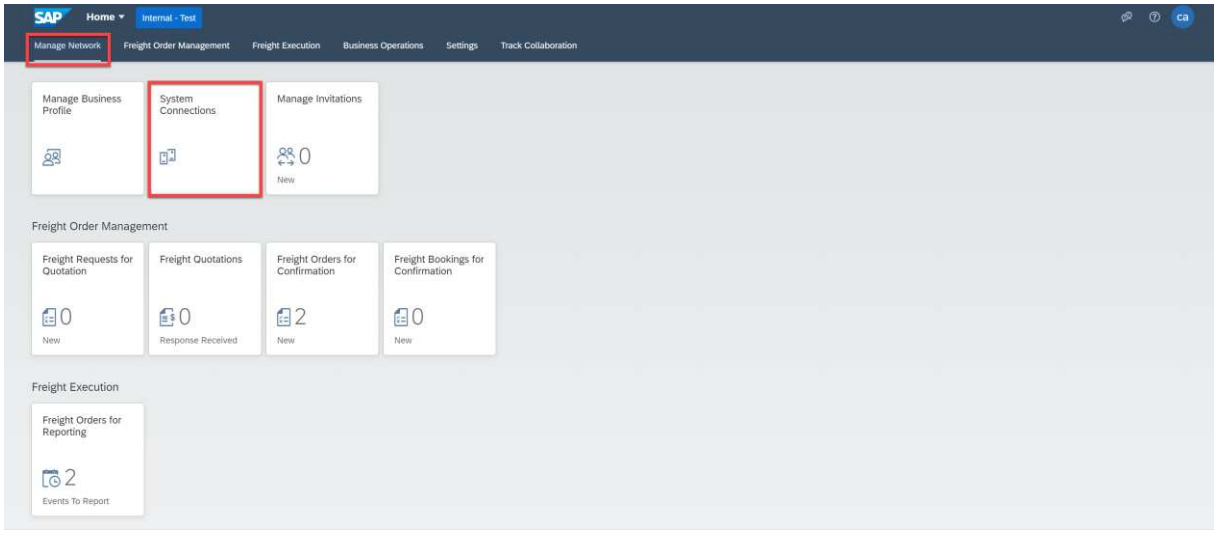
The process flow below illustrates a sequence of events for exchanging ANSI messages which are supported by SAP Business Network Freight Collaboration. The supported message types are also listed in the Introduction section.



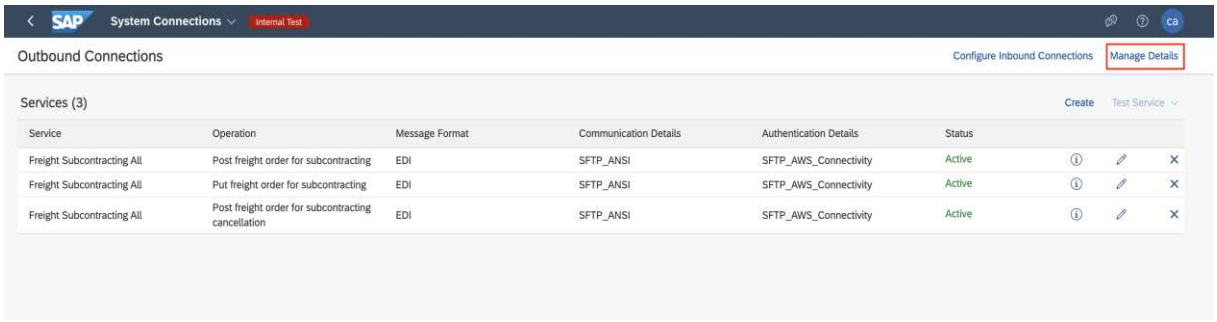
We recommend to read the following blog post *before* starting the integration work: [SAP Business Network for Logistics – Road EDI Integration Tips and Tricks].

STEP 1: ANSI X12 204/300/303/840/864/997 (OUTBOUND)

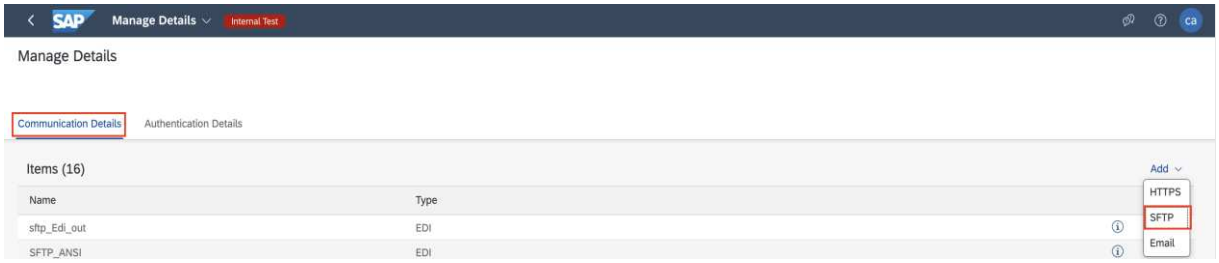
On the *Manage Network* tab of your SAP Fiori launchpad, select the *System Connections* app.



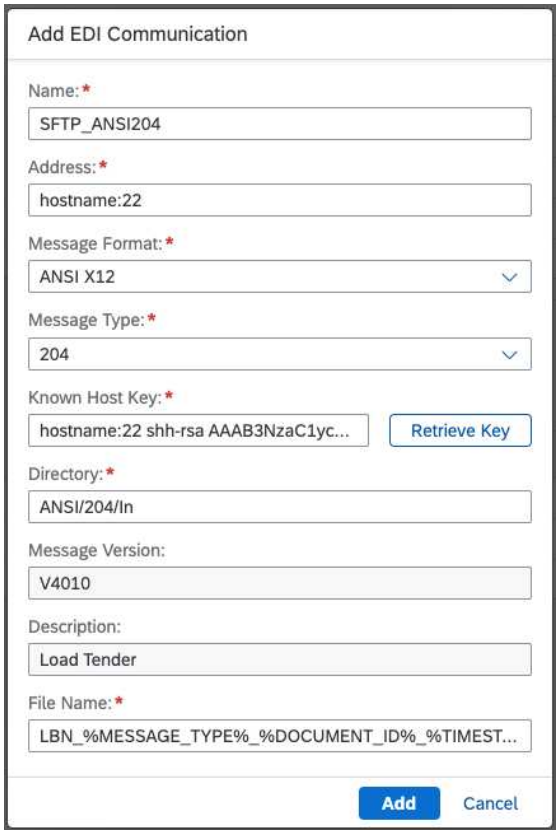
Choose *Manage Details* in the upper right corner.



Under the *Communication Details* tab, choose *Add* and select the SFTP option from the dropdown menu.



Enter the following data in the *Add EDI Communication* window:



For more information, see Specifying the Authentication Details and Base URLs for Your System's Endpoints.

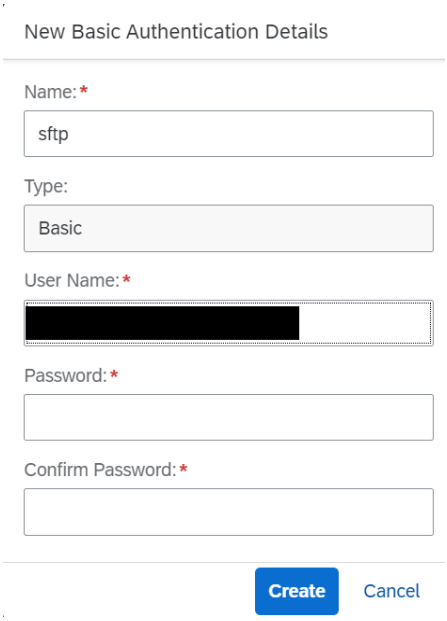
> **Hint**: The file name can consist of particular placeholders that you replace when creating the file name and fixed parts. For example, LBN_%DOCUMENT_ID%_%TIMESTAMP%.edi
>
> Available placeholders:
>
> - %DOCUMENT_ID%
> - %MESSAGE_TYPE%
> - %SHIPPER_LBN_ID%
> - %TIMESTAMP%

In addition, for the EDI 997 message, the placeholder %CONTROL_NUMBER% is available to be used as a unique ID referring to each ST/SE transaction in LBN inbound message containing multiple transactions.

Under the *Authentication Details* tab, choose *Add* and select the BASIC option from the dropdown menu.



Enter the following data in the *New Basic Authentication Details* window:

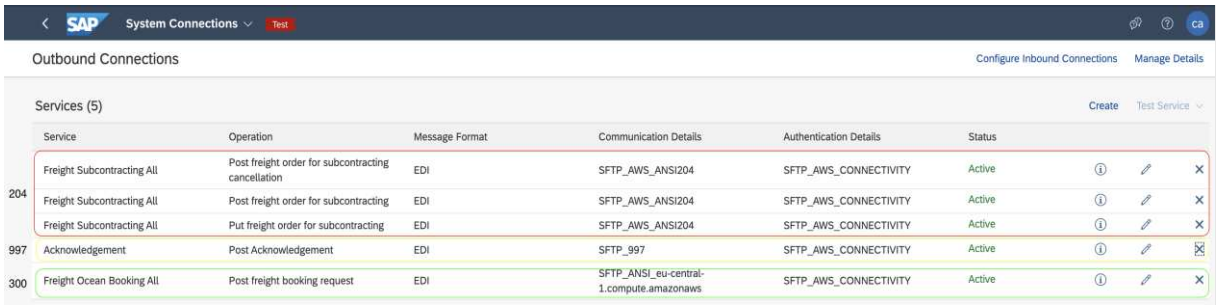


Go back to the entry page of the *System Connections* app and create different outbound connections for the following services and operations:

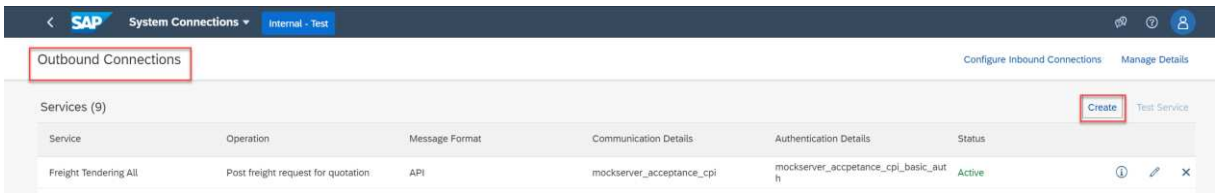
Service	Operation	Corresponding ANSI Message
Freight Subcontracting All	Post freight order for subcontracting cancellation	ANSI 204
	Post freight order for subcontracting	ANSI 204
	Put freight order for subcontracting	ANSI 204
Freight Ocean Booking All	Post freight booking request	ANSI 300
	Post freight booking request cancellation	ANSI 303
Acknowledgement	Post acknowledgement	ANSI 997
Freight Tendering All	Post freight request for quotation	ANSI 840
	Post freight request for quotation cancellation	ANSI 840
	Post freight quotation notification	ANSI 864
Freight Road Tendering FTL	Post freight request for quotation	ANSI 840
	Post freight request for quotation cancellation	ANSI 840

	Post freight quotation notification	ANSI 864
Freight Road Tendering LTL	Post freight request for quotation	ANSI 840
	Post freight request for quotation cancellation	ANSI 840
	Post freight quotation notification	ANSI 864

This will look as follows:



To do so, create a new service:



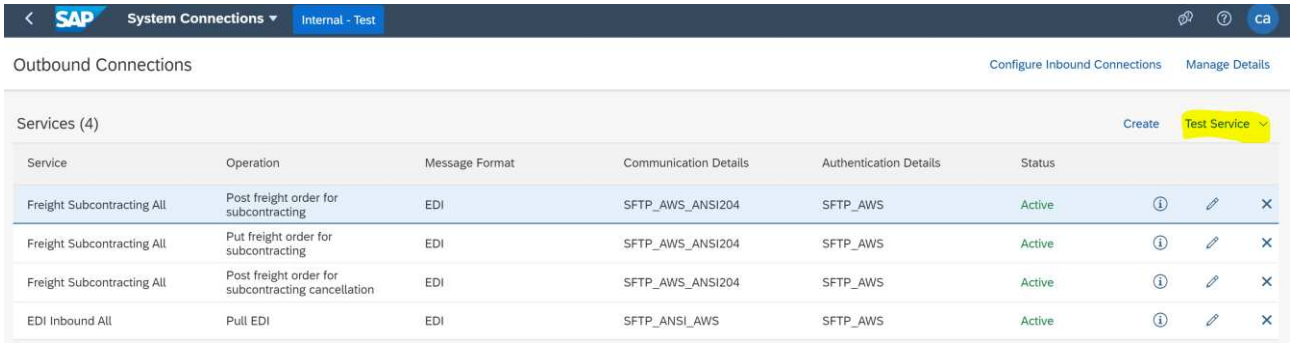
Enter the following data for the new service:



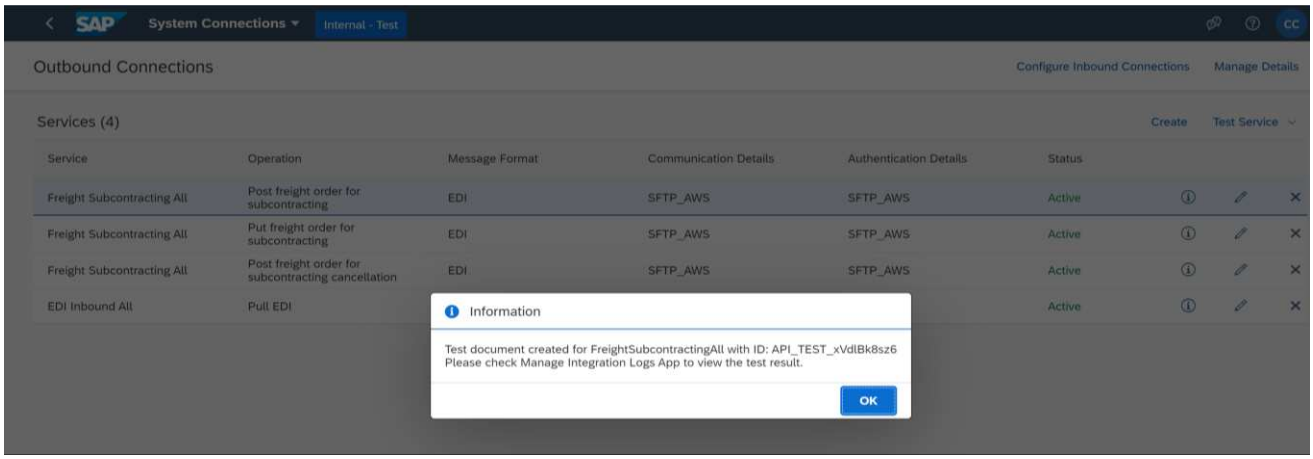
Repeat these steps for each of the following operations for all Services mentioned in the table above.

STEP 2: TEST OUTBOUND CONNECTIVITY

After configuration, you should directly perform a connection test to your SFTP server. To test the service, in the *Services* table, select *Post freight order for subcontracting* and choose *Test Service* in the upper-right corner, then select *With Default Payload*:



The system generates a test freight order and attempts to send a 204 file to your SFTP server. The below window is shown with the test freight order ID:



Using the ID of the created document (for example API_TEST_xVdlBk8sz6), open the *Manage Integration Logs* app to view the test result. Please refer to the chapter OPERATIONAL MONITORING for more details.